

Audio Visualizer: Sizing the LED Loudness Ladder

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Ten LEDs per channel that light in sequence with loudness. No driver IC, just transistors, resistors, and half an LM358 per channel. This is the math behind the resistor values.

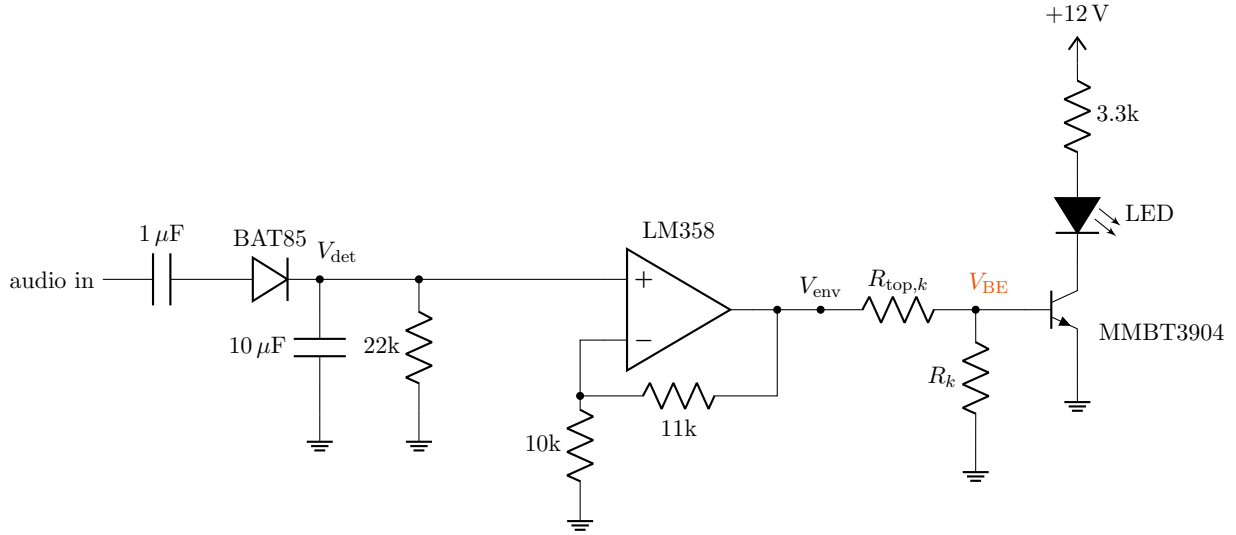


Figure 1: Front end and one of ten ladder stages (one channel of two). The divider pair $R_{\text{top},k}/R_k$ changes from stage to stage. LED k turns on when its base hits V_{BE} ; the threshold V_k is not a node, just the V_{env} value that gets the base there.

Getting a loudness voltage

Audio is AC; the ladder needs a DC voltage that tracks volume. A $1\mu\text{F}$ cap couples the signal in, a Schottky rectifies it, a $10\mu\text{F}$ cap holds the peak, and a $22\text{k}\Omega$ bleed lets it fall ($\tau = 22\text{k}\Omega \times 10\mu\text{F} \approx 0.22\text{s}$). That's V_{det} .

I call $V_{\text{det}} = 5\text{V}$ clipping. That's an educated guess: the speakers can theoretically swing 6V , but the Schottky eats $\sim 0.3\text{V}$ right after the coupling cap and the other parts in series take their share. This display is purely cosmetic, so being off by a little here costs nothing. 5V it was.

Log spacing, because hearing is logarithmic

Equal steps in perceived loudness are equal voltage *ratios*, not equal volts. So the thresholds are spaced in dB, the same call the LM3915 bar driver this circuit replaces makes. 30dB of range, 13

steps, 2.5 dB each:

$$L_k = -30 + 2.5(k - 1) \text{ dB}, \quad V_k = V_{\text{fs}} \cdot 10^{L_k/20}. \quad (1)$$

Filling the 12 V rail

The detector tops out at 5 V but the board runs on 12. An LM358 on a 12 V rail can only swing to about 10.5 V, so I picked the gain to land full scale exactly there: $A = 10.5/5 = 2.1$, built as $1 + 11\text{k}/10\text{k}$. Full scale and clipping are the same voltage on purpose, which makes the top LED the clip light. The op amp also buffers the detector: the ladder pulls its current from the op amp instead of draining the peak-hold cap.

One transistor, three resistors per LED

Each stage is an NPN with a grounded emitter and an LED with its $3.3\text{k}\Omega$ current limiter on the collector. Two more resistors form the base divider that decides when the stage turns on:

$$V_k = V_{\text{BE}} \left(1 + \frac{R_{\text{top}}}{R_k} \right) \implies \frac{R_{\text{top}}}{R_k} = \frac{V_k}{V_{\text{BE}}} - 1. \quad (2)$$

A divider can only divide, so nothing below $V_{\text{BE}} \approx 0.65\text{ V}$ is reachable. Any division under that line simply can't be built. The 13 came from trial and error: with 13 divisions, exactly three (0.33, 0.44, 0.59 V) fall under V_{BE} , and the 10 LEDs I wanted survive. Usable range: -22.5 to 0 dB .

My first pass fixed $R_{\text{top}} = 10\text{ k}\Omega$ everywhere: one resistor to calculate per LED, about a milliamp per divider at full scale. Then the JLCPCB order got real. It's not enough for a value to exist as a standard E24 part; it also has to be in JLCPCB's *basic* catalog, because every unique extended part adds a loading fee to assembly, and several of the 10k-anchored values were extended only. Since the divider only cares about the ratio in Eq. (2), I re-picked both resistors per stage from the basic catalog. Dividers stay in the same roughly-a-milliamp range, and the errors actually improved: worst case fell from 3.8% to 1.4%. The error column is how far the built threshold lands from the ideal V_k ; adjacent thresholds are 33% apart, so turn-on order was never in danger:

LED	L_k	ideal V_k	R_{top}	R_k	threshold	error
1	-22.5 dB	0.787 V	10 k Ω	47 k Ω	0.788 V	+0.1%
2	-20.0 dB	1.050 V	20 k Ω	33 k Ω	1.044 V	-0.6%
3	-17.5 dB	1.400 V	27 k Ω	24 k Ω	1.381 V	-1.4%
4	-15.0 dB	1.867 V	22 k Ω	12 k Ω	1.842 V	-1.4%
5	-12.5 dB	2.490 V	6.8 k Ω	2.4 k Ω	2.492 V	+0.1%
6	-10.0 dB	3.320 V	10 k Ω	2.4 k Ω	3.358 V	+1.1%
7	-7.5 dB	4.428 V	27 k Ω	4.7 k Ω	4.384 V	-1.0%
8	-5.0 dB	5.905 V	22 k Ω	2.7 k Ω	5.946 V	+0.7%
9	-2.5 dB	7.874 V	30 k Ω	2.7 k Ω	7.872 V	-0.0%
10	0.0 dB	10.500 V	15 k Ω	1 k Ω	10.400 V	-1.0%

Collector resistors: $R_C = (12 - 2.0 - 0.2)\text{ V}/3\text{ mA} \approx 3.3\text{ k}\Omega$ per LED. The board carries two identical channels, left and right.

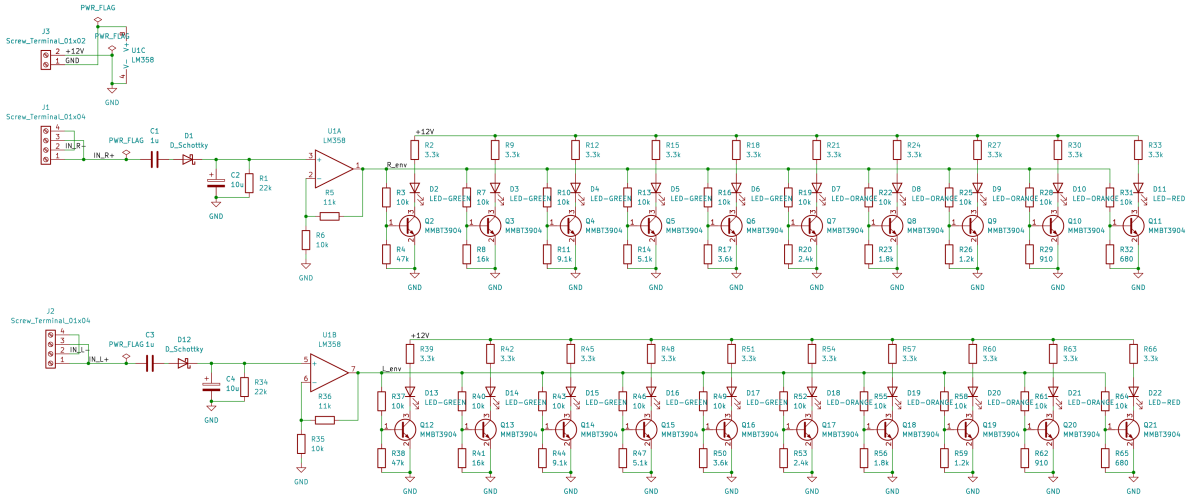


Figure 2: Full schematic: both channels, each half of the LM358 driving its own ten-stage ladder.

